

ABM BUSINESS LOGIC BIBLE

§10 v2 — REALITY CALIBRATION & INSTITUTIONAL LEARNING ENGINE

10 Iterations · 10 Architectural Layers · 26 New Rules · 4 EPIS Integration Decisions · 1 Data-Model Implication

§10 v1 specified event capture, sentiment classification, and engagement scoring. §10 v2 reframes the engine itself. **§10 is not an engagement-tracking system. It is a belief-correction system.** Everything before §10 is belief: signals are belief, scores are belief, narratives are belief, strategies are belief, sequences are belief. §10 is where reality votes. Its purpose is not to record what happened — it is to continuously compare the engine's beliefs against observed reality, distinguish behavior from intent, detect meaningful absences, account for sensor imperfections, prevent contaminated learning, and convert transient events into permanent organizational memory.

Iterations	10	Layers	10
New rules	26	EPIS integrations	4 explicit

FOUR INTEGRATION DECISIONS — NOT A SECOND EPISTEMOLOGY

Intent Confidence + Sensor Reliability ARE EPIS-RCM, applied to engagement events. Not new scores — the existing RCM stamp (EPIS-RCM-01) extended to cover the engagement-event subtype of SIGNAL, with source incentive-bias (EPIS-SRC-01) reused as the sensor-reliability axis.

Belief-correction ('which prior belief did this strengthen or weaken') IS EPIS-RCM-03's falsifiability requirement, applied retrospectively to closed engagement events rather than prospectively to open beliefs.

Cause Attribution + Counterfactual Estimate are ONE mechanism: a cause-attribution distribution over named candidates (outreach / relationship / narrative / market event / competitor action / unknown), with the counterfactual baseline as one explicit candidate — not two separate, driftable rule families.

Organizational Memory is §10's CONTRIBUTION to Tier 6 Strategic Memory (§2 OrganizationalDNA, MarketNarrative, CounterfactualRecord, FailurePatternRecord) — not a competing memory system.

THE TEN LAYERS — §10 v2 ARCHITECTURE MAP

#	Layer	From iter.	What it adds
1	Observation vs Intent (Epist)	1	Behavior ≠ intent; Intent Confidence is EPIS-RCM applied to events
2	Absence of Signal (Null)	2	Silence relative to an EXPECTED pattern is itself a negative signal event
3	Sensor Reliability (Sensor)	3	The engine observes proxies, not reality; source reliability weights learning
4	Cause Attribution (Learn/CF)	4, 7	Cause-attribution distribution + counterfactual baseline, one mechanism
5	Belief Correction (Truth-1)	5	Every event answers: which prior belief did this strengthen/weaken?
6	Reality vs Perception (Real)	6	Engagement measures perception; never let it alone move Opportunity score
7	Multi-Stakeholder Truth (Truth-2)	8	Organizational Ambiguity State — flagged as a §2 data-model implication
8	Longitudinal Learning (Long)	9	Trajectory beats snapshot; rising-40 beats falling-80
9	Organizational Memory (Mem)	10	§10's contribution to Tier 6 — not a new memory system
10	Synthesis (Truth-deepest)	5+10	§10's true output is institutional learning, not event records

LAYER 1 — OBSERVATION vs INTENT (EPIST)

An email click could mean genuine interest, an accidental tap, curiosity, competitor research, or an assistant screening vendor mail. §10 v1 observes behavior. It does not, on its own, observe motive. The fix is not a new score — it's applying the RCM stamp the architecture already has to this specific evidence category.

Rule ID	Phase	Specification
ENG-EPIST-01	P1	Every engagement event carries an RCM stamp (EPIS-RCM-01) with an Intent Confidence component distinct from Event Confidence (did this event technically occur vs. what does it imply about motive). Meeting Booked carries very high Intent Confidence (externally verified, unambiguous); a single Email Open carries very low Intent Confidence (multiple non-intent explanations exist).
ENG-EPIST-02	P1	Observed Behavior and Inferred Intent are stored as separate fields on every engagement record, never collapsed into one. Downstream consumers (§4 scoring, §7 v2 Trust Accumulation) read Inferred Intent, never raw Observed Behavior directly — preventing the over-learning that comes from treating every click as equally meaningful.

LAYER 2 — ABSENCE OF SIGNAL (NULL)

Rule ID	Phase	Specification
ENG-NULL-01	P2	Every contact with sufficient history (per §10 v1 ENG-ARCH archetype minimum-evidence floor) gets an Expected Engagement Pattern : typical reply latency, typical engagement cadence, derived from their own history.
ENG-NULL-02	P2	A material deviation from the Expected Engagement Pattern (e.g. a contact who normally replies within 3 days going silent for 30) generates a Negative Signal Event — silence becomes a first-class logged event, not merely an absence of logged events.
ENG-NULL-03	P1	Negative Signal Events are interpreted through Engagement Archetype (§10 v1 ENG-ARCH, §9 v2 OUT-BEH) exactly as ENG-SENT-06/ENG-SILENCE already require — this layer generates the event; archetype interpretation (already specified) consumes it. No new interpretation mechanism.

LAYER 3 — SENSOR RELIABILITY (SENSOR)

Open pixels get blocked or auto-fire on security gateway scans. Forwarded emails create false activity. Browser prefetch triggers phantom clicks. The engine observes proxies for reality, not reality itself — and different proxies have very different reliability.

Rule ID	Phase	Specification
ENG-SENSOR-01	P1	Every event SOURCE (not just each individual event) carries a Reliability tier, reusing EPIS-SRC-01's reliability/incentive-bias separation: Meeting Booked → Very High (externally verified by calendar/CRM); Reply → High (requires actual human action, though sentiment can still be misread); Click → Medium (link-prefetch and security-scan false positives are common); Open → Low (pixel-blocking and auto-fire are common).
ENG-SENSOR-02	P1	§13 Analytics and any learning loop (§4 SCORE-RECAL, §7 v2 prompt optimization) weight observations by source reliability — a Click-sourced signal contributes less to a learned conclusion than a Reply-sourced signal of equal apparent sentiment. This rule is what makes Layer 1's Intent Confidence and this layer's source reliability operate as one coherent weighting system, not two.

LAYER 4 — CAUSE ATTRIBUTION & COUNTERFACTUAL BASELINE (LEARN/CF)

A bank already evaluating API platforms replies to Decimal's email. Did the email work, or would the reply have happened anyway? A consultant's private introduction precedes a reply — did outreach succeed, or did the relationship? Iterations 4 and 7 are the same problem from two angles; specified as one mechanism.

Rule ID	Phase	Specification
ENG-CAUSE-01	P2	Every significant engagement outcome (positive reply, meeting booked) receives a Cause Attribution Distribution over named candidates: outreach, pre-existing relationship, narrative/market timing, competitor action, unknown. This is a distribution (e.g. outreach 40% / relationship 35% / market timing 25%), not a single confident label — consistent with EPIS-RCM-03's anti-false-confidence discipline.
ENG-CAUSE-02	P2	One named candidate in every Cause Attribution Distribution is always the Counterfactual Baseline: the estimated probability this outcome would have occurred with no Decimal outreach at all. Where available, COLD/WATCHING accounts with comparable profiles and no active outreach serve as a reference cohort for calibrating this baseline (V2 §2.1 account states already include WATCHING, which can serve this role without a new entity).
ENG-CAUSE-03	P1	Learning systems (§4 SCORE-RECAL, §7 v2 prompt optimization, §8 v2 Trust Preservation Score) consume Cause-Attributed outcomes, weighted by the non-counterfactual share of the distribution — NOT raw outcomes. An outcome attributed 70% to counterfactual baseline contributes only 30% of its raw weight to any 'this worked' conclusion.

LAYER 5 — BELIEF CORRECTION (TRUTH-1)

The deepest discovery from the first five iterations: §10's purpose is not measuring success, not even reducing uncertainty — it is correcting the engine's model of reality. This is EPIS-RCM-03 (every confidence figure links to the evidence that produced it AND the evidence that would lower it) applied retrospectively: an event has now arrived that IS that lowering or raising evidence.

Rule ID	Phase	Specification
ENG-TRUTH-01	P1	Every significant engagement event is tagged with the specific prior belief(s) it bears on — named explicitly: a timing hypothesis (§5 v3 Patience layer), a persona hypothesis (§7 Agent 3), a narrative hypothesis (§7 v2 AI-NARR), or a trust hypothesis (§7 v2 AI-TRUST). An event with no identifiable prior belief it bears on is logged but does not feed any update.
ENG-TRUTH-02	P1	§10's defined purpose is Model Correction, not Event Recording — this is the explicit architectural test for §10 rules going forward (mirrors §5 v3 TIER-ATTN-003 and §6 v3's information-economics test as the same kind of binding self-test). A proposed §10 rule that does not update some specific upstream belief is not a §10 rule.

LAYER 6 — REALITY vs PERCEPTION (REAL)

Rule ID	Phase	Specification
ENG-REAL-01	P1	Engagement Reality (the observed response) is explicitly separated from Opportunity Reality (the inferred underlying buying state, per §4's Effective Opportunity / SIG-VSAT displacement / SIG-HYP). Engagement score alone NEVER directly sets or moves Opportunity score — it is one weighted input among the §4 dimensions, gated through Layer 4's cause attribution first.
ENG-REAL-02	P1	Positive engagement with no supporting independent signal (no SIG-NEWS/SIG-REG/SIG-VENDOR corroboration) creates Curiosity — a flagged hypothesis for further verification — not Certainty. This binds via EPIS-COV-02 (coverage caps confidence): an account whose only positive evidence is a single enthusiastic reply has, by definition, low Intelligence Coverage, and the engine's confidence is capped accordingly regardless of how positive the reply sounds.

LAYER 7 — MULTI-STAKEHOLDER TRUTH (TRUTH-2)

CTO enthusiastic, Compliance resistant, Procurement silent. None of them alone represents account-level truth; the organization contains multiple simultaneous, possibly conflicting realities. This extends §9 v2's (dormant) Stakeholder State Map from a per-contact telemetry record into an explicit account-level synthesis.

Rule ID	Phase	Specification
ENG-TRUTH-03	P2	A Stakeholder Truth Map extends the §9 v2/§10 v1 Stakeholder State Map with: support level, resistance level, influence level (per §3 SIG-POWER), and confidence level per committee member — a richer per-contact record, still observational only pending OPEN-Q9.3.
ENG-TRUTH-04	P2	Account-level truth is derived from WEIGHTED stakeholder aggregation (weighted by each member's SIG-POWER influence level), not from any single contact's engagement — this is read-only synthesis for human/analytics consumption (§11, §13), not a sequence-execution input, consistent with OPEN-Q9.3's unresolved status.
ENG-TRUTH-05	P2	Conflicting stakeholder signals (e.g. CTO supportive AND Compliance resistant, both at high confidence) produce an Organizational Ambiguity State on the account.

DATA-MODEL IMPLICATION — ORGANIZATIONAL AMBIGUITY STATE BELONGS IN §2

Iteration 8 proposes Organizational Ambiguity State as a 'first-class entity.' That is a §2 (Master Entity Model) decision, not something a §10 document can quietly create. §10 v2 specifies the STATE and its derivation rule (ENG-TRUTH-05) but flags the entity itself as a **pending §2 addendum** — to be formally added to the entity model (likely Tier 3 Decision, alongside Hypothesis and InfluenceMap) in a future §2 revision, rather than introduced informally here.

LAYER 8 — LONGITUDINAL RELATIONSHIP LEARNING (LONG)

Trust moving 20→40→60→80 and trust moving 80→60→40→20 average to the same value but predict opposite futures. This is §9 v2 OUT-COMPREL's Relationship Momentum (Layer 10 there), generalized here as the engagement-tracking-side trajectory record that Momentum reads from.

Rule ID	Phase	Specification
ENG-LONG-01	P2	Every contact maintains a Relationship Trajectory — the time series of Trust Accumulation Score, Engagement Score, and Intent Confidence values, not just current snapshots. This is the data record; §9 v2 OUT-COMPREL-01 (Relationship Momentum) is the rate-of-change computation derived FROM this trajectory — one record, one derived metric, not two parallel histories.
ENG-LONG-02	P1	Trend direction is read alongside current value in any downstream decision that uses Trust Accumulation or Engagement Score: a rising-40 and a falling-80 are NOT treated as equivalent inputs even when comparing two contacts at similar current values. This generalizes §9 v2 OUT-COMPREL-02 from sequence-pause decisions to all downstream consumers (§4, §5, §11).

LAYER 9 — ORGANIZATIONAL MEMORY (MEM)

Rule ID	Phase	Specification
ENG-MEM-01	P2	Every significant engagement outcome (positive reply, meeting booked, or a clear disqualification) is packaged as a Learning Artifact — context, action, response, outcome, cause attribution, and lesson — and submitted to Tier 6 Strategic Memory (§2: specifically feeding MarketNarrative and CounterfactualRecord) rather than existing only as a row in the §10 event log.
ENG-MEM-02	P1	Patterns extracted from Learning Artifacts (e.g. 'this narrative angle underperforms with Compliance personas at Islamic banks') are stored as reusable organizational knowledge in Tier 6, NOT attributed to or siloed by the original account_owner — the knowledge survives personnel turnover by construction, because it lives in Tier 6, not in any individual's notes.
ENG-MEM-03	P1	§10's defined output, per ENG-TRUTH-02's architectural test, includes its Tier 6 contribution as a first-class deliverable — a §10 implementation that logs events but never produces Learning Artifacts is incomplete, regardless of how complete its event-capture coverage is.

§10 v2 RULE ROLL-UP & PLACEMENT STATUS

Layer	Rules	Phase	Cross-section binding
1 Epist	ENG-EPIST-01..02	P1	EPIS-RCM-01 (reused, not duplicated)
2 Null	ENG-NULL-01..03	P1 / P2	§9 v1 ENG-SILENCE, §10 v1 ENG-ARCH (consumes, doesn't replace)
3 Sensor	ENG-SENSOR-01..02	P1	EPIS-SRC-01 (reused), §13 learning loops
4 Learn/CF	ENG-CAUSE-01..03	P1 / P2	EPIS-RCM-03, V2 §2.1 WATCHING-state reference cohort
5 Truth-1	ENG-TRUTH-01..02	P1	EPIS-RCM-03 retrospective application; the §10 architectural test
6 Real	ENG-REAL-01..02	P1	§4 Effective Opportunity, EPIS-COV-02 coverage cap
7 Truth-2	ENG-TRUTH-03..05	P2	SIG-POWER, §2 data-model implication (pending addendum), OPEN-Q9.3
8 Long	ENG-LONG-01..02	P1 / P2	§9 v2 OUT-COMPREL (one record, shared derived metric)
9 Mem	ENG-MEM-01..03	P1 / P2	Tier 6 Strategic Memory (§2), not a new memory system

§10 v2 COMPLETE — WHAT IT ESTABLISHED

§10 is no longer an engagement-tracking system. It is the engine's **Reality Calibration and Institutional Learning Engine**: it separates behavior from intent, treats meaningful silence as a logged event, weights evidence by sensor reliability, attributes outcomes across causal candidates including a counterfactual baseline, asks of every event which prior belief it updates, keeps engagement and opportunity as distinct realities, synthesizes (but doesn't yet act on) multi-stakeholder truth, tracks trajectory over snapshot, and converts transient events into permanent Tier 6 organizational memory. 26 new rules, with four explicit decisions that bind every new layer to EPIS and Tier 6 rather than building parallel mechanisms, plus one flagged §2 data-model implication (Organizational Ambiguity State) that requires a future entity-model addendum rather than being silently introduced here.

The architectural test going forward: a §10 rule that does not update some specific upstream belief, and does not eventually feed Tier 6, is not a §10 rule.

PLACEMENT MAP UPDATE

§10 status remains Complete-with-Opens (OPEN-Q10.1, OPEN-Q10.2 from §10 v1 unchanged). §2 (Master Entity Model) gains a NOTED PENDING ADDENDUM: Organizational Ambiguity State, to be formally specified when §2 is next revised — tracked here so it isn't lost, not yet built.

ON THE HORIZON

§11 Human Handoff Engine — per the prior assessment, this opens a genuinely new domain (human-AI collaboration, escalation economics, expert judgment transfer, decision accountability, knowledge packaging) rather than continuing to deepen engagement tracking. §11 inherits §10 v2's Cause Attribution, Belief Correction tagging, and Learning Artifacts as the substance of the handoff package's context.